KickIT: Football Scouting Database System

Presented By:

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Problem Domain & Motivation

The Scouting Challenge

- Football clubs track 1000+ players across 50+ countries simultaneously
- Data scattered across spreadsheets, emails, and manual reports
- Critical decisions (multi-million dollar transfers) made with incomplete information
- No centralized system to track scout performance or market trends

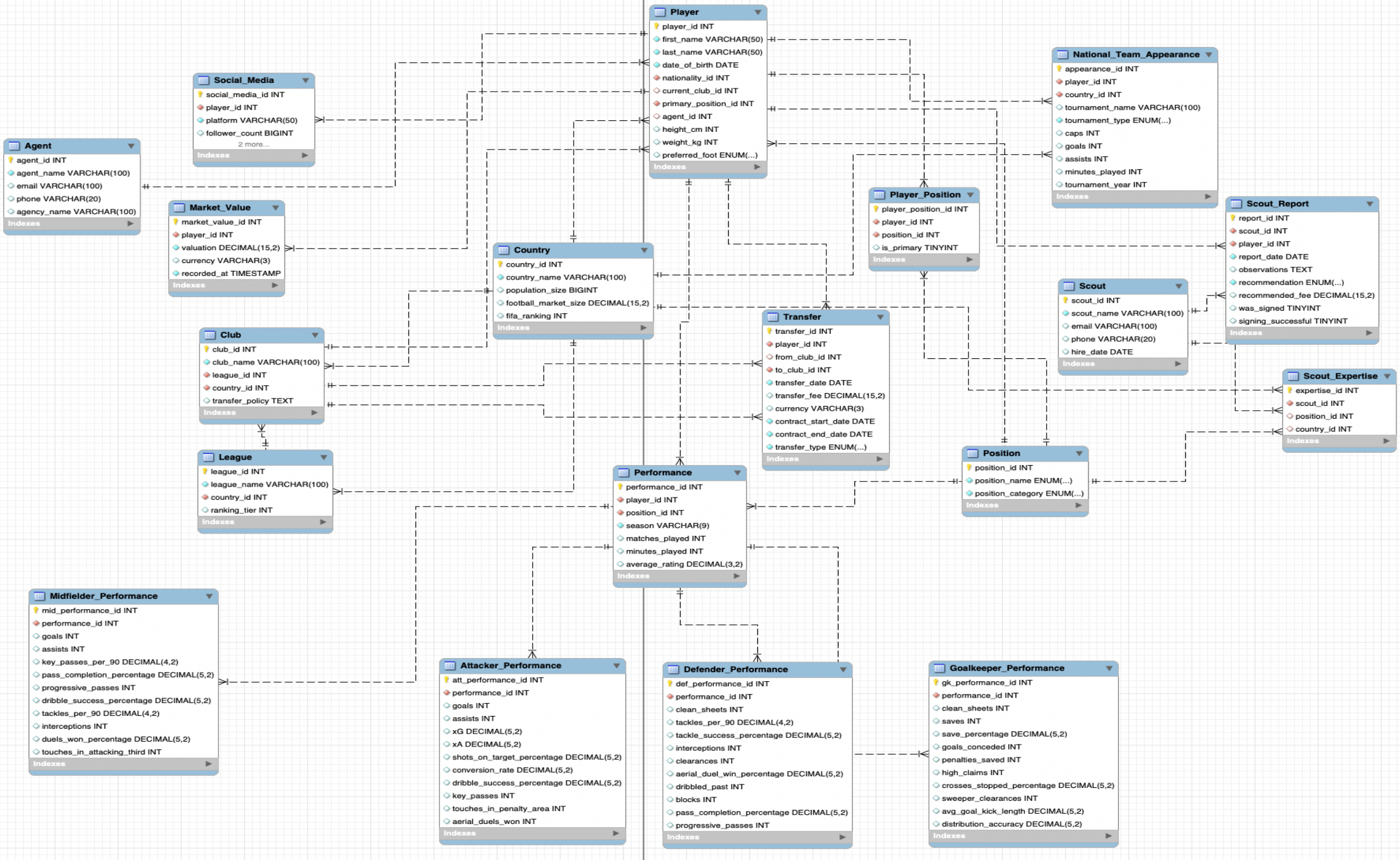
Database Solution Needed

- Centralized data repository for player tracking
- Support complex queries for talent identification
- Maintain historical records for trend analysis
- Ensure data integrity across interconnected entities

Entity-Relationship Overview

Primary Entities

1. **Player** (player demographics, physical attributes, current club, nationality, position)
2. **Club** (team information, league, country, transfer policies)
3. **Scout and Scout expertise** (scout details, assignments, expertise areas by position and country)
4. **Performance** (match statistics, ratings, metrics by position and season)
5. **Transfer** (fees, dates, contract details, contract expiration dates)
6. **Position** (player positions, primary/secondary roles)
7. **Country** (country name, leagues, population size, FIFA ranking)
8. **League** (league name, country, ranking/tier level)
9. **Scout Report** (detailed observations, recommendations, player signing outcomes)
10. **Agent** (agent details, agency name)
11. **National Team Appearances** (tournaments, caps, major competition participation)
12. **Social Media** (platform, follower count, growth rate, timestamp)
13. **Market Value** (player valuation, timestamp, currency)


Player
player_id INT
first_name VARCHAR(50)
last_name VARCHAR(50)
date_of_birth DATE
nationality_id INT
current_club_id INT
primary_position_id INT
agent_id INT
height_cm INT
weight_kg INT
preferred_foot ENUM(...)
Indexes
Social_Media
social_media_id INT
player_id INT
platform VARCHAR(50)
follower_count BIGINT
2 more...
Indexes
Agent
agent_id INT
agent_name VARCHAR(100)
email VARCHAR(100)
phone VARCHAR(20)
agency_name VARCHAR(100)
Indexes
Market_Value
market_value_id INT
player_id INT
valuation DECIMAL(15,2)
currency VARCHAR(3)
recorded_at TIMESTAMP
Indexes
National_Team_Appearance
appearance_id INT
player_id INT
country_id INT
tournament_name VARCHAR(100)
tournament_type ENUM(...)
caps INT
goals INT
assists INT
minutes_played INT
tournament_year INT
Indexes
Player_Position
player_position_id INT
player_id INT
position_id INT
is_primary TINYINT
Indexes
Scout_Report
report_id INT
scout_id INT
player_id INT
report_date DATE
observations TEXT
recommendation ENUM(...)
recommended_fee DECIMAL(15,2)
was_signed TINYINT
signing_successful TINYINT
Indexes
Country
country_id INT
country_name VARCHAR(100)
population_size BIGINT
football_market_size DECIMAL(15,2)
fifa_ranking INT
Indexes
Scout
scout_id INT
scout_name VARCHAR(100)
email VARCHAR(100)
phone VARCHAR(20)
hire_date DATE
Indexes
Transfer
transfer_id INT
player_id INT
from_club_id INT
to_club_id INT
transfer_date DATE
transfer_fee DECIMAL(15,2)
currency VARCHAR(3)
contract_start_date DATE
contract_end_date DATE
transfer_type ENUM(...)
Indexes
Club
club_id INT
club_name VARCHAR(100)
league_id INT
country_id INT
transfer_policy TEXT
Indexes
Scout_Expertise
expertise_id INT
scout_id INT
position_id INT
country_id INT
Indexes
League
league_id INT
league_name VARCHAR(100)
country_id INT
ranking_tier INT
Indexes
Position
position_id INT
position_name ENUM(...)
position_category ENUM(...)
Indexes
Performance
performance_id INT
player_id INT
position_id INT
season VARCHAR(9)
matches_played INT
minutes_played INT
average_rating DECIMAL(3,2)
Indexes
Midfielder_Performance
mid_performance_id INT
performance_id INT
goals INT
assists INT
key_passes_per_90 DECIMAL(4,2)
pass_completion_percentage DECIMAL(5,2)
progressive_passes INT
dribble_success_percentage DECIMAL(5,2)
tackles_per_90 DECIMAL(4,2)
interceptions INT
duels_won_percentage DECIMAL(5,2)
touches_in_attacking_third INT
Indexes
Attacker_Performance
att_performance_id INT
performance_id INT
goals INT
assists INT
xG DECIMAL(5,2)
xA DECIMAL(5,2)
shots_on_target_percentage DECIMAL(5,2)
conversion_rate DECIMAL(5,2)
dribble_success_percentage DECIMAL(5,2)
key_passes INT
touches_in_penalty_area INT
aerial_duels_won INT
Indexes
Defender_Performance
def_performance_id INT
performance_id INT
clean_sheets INT
tackles_per_90 DECIMAL(4,2)
tackle_success_percentage DECIMAL(5,2)
interceptions INT
clearances INT
aerial_duel_win_percentage DECIMAL(5,2)
dribbled_past INT
blocks INT
pass_completion_percentage DECIMAL(5,2)
progressive_passes INT
Indexes
Goalkeeper_Performance
gk_performance_id INT
performance_id INT
clean_sheets INT
saves INT
save_percentage DECIMAL(5,2)
goals_conceded INT
penalties_saved INT
high_claims INT
crosses_stopped_percentage DECIMAL(5,2)
sweeper_clearances INT
avg_goal_kick_length DECIMAL(5,2)
distribution_accuracy DECIMAL(5,2)
Indexes

1.Multi-Position Versatility Report

Description: List all players who have played in three or more different positions with an average performance rating above 7.5 in each position.

player_id	player_name	age	nationality	current_club	positions_played	positions	overall_avg_rating	min_position_rating	max_position_rating
2	Luca Vandenberg	24	Belgium	Club Brugge	3	DM, CM, CAM	7.90	7.70	8.10
1	Marcus Silva	23	Brazil	Flamengo	3	RW, ST, CF	7.87	7.60	8.20

```sql
SELECT
   p.player_id,
   CONCAT(p.first_name, ' ', p.last_name) AS player_name,
   TIMESTAMPDIFF(YEAR, p.date_of_birth, CURDATE()) AS age,
   c.country_name AS nationality,
   cl.club_name AS current_club,
   COUNT(DISTINCT perf.position_id) AS positions_played,
   GROUP_CONCAT(DISTINCT pos.position_name ORDER BY pos.position_name SEPARATOR ', ') AS positions,
   ROUND(AVG(perf.average_rating), 2) AS overall_avg_rating,
   ROUND(MIN(perf.average_rating), 2) AS min_position_rating,
   ROUND(MAX(perf.average_rating), 2) AS max_position_rating
FROM Player p
JOIN Country c ON p.nationality_id = c.country_id
LEFT JOIN Club cl ON p.current_club_id = cl.club_id
JOIN Performance perf ON p.player_id = perf.player_id
JOIN Position pos ON perf.position_id = pos.position_id
WHERE perf.average_rating > 7.5
GROUP BY p.player_id, p.first_name, p.last_name, p.date_of_birth, c.country_name, cl.club_name
HAVING COUNT(DISTINCT perf.position_id) >= 3
ORDER BY positions_played DESC, overall_avg_rating DESC;
```

2. Emerging Talent vs. Established Player Comparison

Description: Compare players under 23 years old whose performance metrics (goals, assists, defensive actions) have improved in the last year against established players (25+ years) in the same position and league.

age_category	position_category	league_name	player_count	avg_age	avg_rating	avg_goals	avg_assists	avg_tackles_per_90
Established	Goalkeeper	Bundesliga	1	26.0	8.20	0.0	0.0	0.00
Emerging	Defender	Croatian First League	1	22.0	8.00	0.0	0.0	3.20
Emerging	Defender	Eredivisie	1	22.0	7.8 8.00	0.0	0.0	3.50
Emerging	Defender	Premier League	1	22.0	7.70	0.0	0.0	2.80
Established	Midfielder	J1 League	1	25.0	7.80	4.0	7.0	2.90
Established	Midfielder	La Liga	1	25.0	8.60	9.0	16.0	2.10
Established	Midfielder	Ligue 1	1	26.0	8.30	10.0	8.0	0.00
Emerging	Attacker	Premier League	2	21.0	8.10	15.0	9.0	0.00
Established	Attacker	Primeira Liga	1	27.0	8.50	26.0	14.0	0.00

```
WITH PlayerAgeCategory AS (
  SELECT
    p.player_id,
    CONCAT(p.first_name, ' ', p.last_name) AS player_name,
    TIMESTAMPDIFF(YEAR, p.date_of_birth, CURDATE()) AS age,
    CASE
      WHEN TIMESTAMPDIFF(YEAR, p.date_of_birth, CURDATE()) < 23 THEN
'Emerging'
      WHEN TIMESTAMPDIFF(YEAR, p.date_of_birth, CURDATE()) >= 25 THEN
'Established'
      ELSE 'Mid-Career'
    END AS age_category,
    pos.position_name,
    pos.position_category,
    l.league_name,
    perf.season,
    perf.average_rating,
    COALESCE(mp.goals, ap.goals, 0) AS goals,
    COALESCE(mp.assists, ap.assists, 0) AS assists,
    COALESCE(dp.tackles_per_90, mp.tackles_per_90, 0) AS tackles_per_90,
    COALESCE(dp.interceptions, mp.interceptions, 0) AS interceptions
  FROM Player p
  JOIN Club cl ON p.current_club_id = cl.club_id
  JOIN League l ON cl.league_id = l.league_id
  JOIN Performance perf ON p.player_id = perf.player_id
  JOIN Position pos ON perf.position_id = pos.position_id
  LEFT JOIN Midfielder_Performance mp ON perf.performance_id =
mp.performance_id
  LEFT JOIN Attacker_Performance ap ON perf.performance_id =
ap.performance_id
  LEFT JOIN Defender_Performance dp ON perf.performance_id =
dp.performance_id
  WHERE perf.season = '2023-2024'
)
```

```
SELECT
  age_category,
  position_category,
  league_name,
  COUNT(*) AS player_count,
  ROUND(AVG(age), 1) AS avg_age,
  ROUND(AVG(average_rating), 2) AS avg_rating,
  ROUND(AVG(goals), 1) AS avg_goals,
  ROUND(AVG(assists), 1) AS avg_assists,
  ROUND(AVG(tackles_per_90), 2) AS avg_tackles_per_90
FROM PlayerAgeCategory
WHERE age_category IN ('Emerging', 'Established')
GROUP BY age_category, position_category, league_name
ORDER BY position_category, league_name, age_category;
```

3. Undervalued Market Opportunities

- **Description:** List players whose average performance rating exceeds 7.0 but play in leagues ranked outside the top 5 European leagues and have transfer values below $30 million.

player_id	player_name	age	primary_position	club_name	league_name	ranking_tier	league_country	avg_rating	current_market_value_eur	value_in_millions	social_media_followers
5	Nikola Petrovic	22	CB	Dinamo Zagreb	Croatian First League	3	Croatia	8.00	28000000.00	28.00	180000
9	Mateo Gonzalez	24	CDM	América	Liga MX	2	Mexico	7.90	18000000.00	18.00	280000
2	Luca Vandenberg	24	CM	Club Brugge	Belgian Pro League	2	Belgium	7.90	25000000.00	25.00	320000
12	André Oliveira	22	LB	Ajax Amsterdam	Eredivisie	2	Netherlands	7.80	12000000.00	12.00	NULL
6	Takeshi Yamamoto	25	DM	Yokohama F. Marinos	J1 League	2	Japan	7.70	22000000.00	22.00	450000

```
SELECT
   p.player_id,
   CONCAT(p.first_name, ' ', p.last_name) AS player_name,
   TIMESTAMPDIFF(YEAR, p.date_of_birth, CURDATE()) AS age,
   pos.position_name AS primary_position,
   cl.club_name,
   l.league_name,
   l.ranking_tier,
   c.country_name AS league_country,
   ROUND(AVG(perf.average_rating), 2) AS avg_rating,
   ROUND(mv.valuation, 2) AS current_market_value_eur,
   ROUND(mv.valuation / 1000000, 2) AS value_in_millions,
   sm.follower_count AS social_media_followers
FROM Player p
JOIN Club cl ON p.current_club_id = cl.club_id
JOIN League l ON cl.league_id = l.league_id
JOIN Country c ON l.country_id = c.country_id
JOIN Position pos ON p.primary_position_id = pos.position_id
JOIN Performance perf ON p.player_id = perf.player_id
JOIN Market_Value mv ON p.player_id = mv.player_id
LEFT JOIN (
SELECT player_id, SUM(follower_count) AS follower_count
   FROM Social_Media
   WHERE recorded_at = (SELECT MAX(recorded_at)
FROMSocial_Media)
   GROUP BY player_id
) sm ON p.player_id = sm.player_id
WHERE l.ranking_tier > 1
  AND mv.recorded_at = (
     SELECT MAX(recorded_at)
     FROM Market_Value
     WHERE player_id = p.player_id
  )
  AND mv.valuation < 30000000
GROUP
BY p.player_id, p.first_name, p.last_name, p.date_of_birth,pos.
position_name,
       cl.club_name, l.league_name, l.ranking_tier, c.country_nam
e,mv.valuation, sm.follower_count
HAVING avg_rating > 7.0
ORDER BY avg_rating DESC, value_in_millions ASC;
```

4. Scout Specialization Effectiveness

- **Description:** For each scout, show which positions and countries they've evaluated, along with the percentage of their recommended players who were actually signed and later had successful performance ratings (above 7.0).

scout_id	scout_name	expertise_positions	expertise_countries	total_reports	sign_recommendations	players_signed	successful_signings	signing_conversion_rate_pct	success_rate_pct	avg_rating_of_signed_players
4	Sophie Leblanc	CAM, RW	Belgium, France	1	3	3	3	100.00	100.00	8.30
5	Marco Rossi	LW, RW	Italy, Portugal	1	3	3	3	100.00	100.00	8.50
6	João Santos	NULL	Argentina, Brazil	1	4	4	4	100.00	100.00	8.10
1	Carlos Rodriguez	ST, CF	Argentina, Spain	2	9	3	3	33.33	100.00	8.60
3	Hans Mueller	DM, CDM, CM	Germany	1	6	0	0	0.00	NULL	NULL
8	Dragan Kovač	CB	Croatia	1	2	0	0	0.00	NULL	NULL
11	Thomas Anderson	CM, CAM	England	1	2	0	0	0.00	NULL	NULL
2	Emma Thompson	CB, RB	England, Netherlands	1	0	0	0	NULL	NULL	NULL
7	Pieter van Dijk	LB, LWB, CM	Netherlands	1	0	0	0	NULL	NULL	NULL
9	Hiroshi Tanaka	DM	Japan	1	0	0	0	NULL	NULL	NULL
10	Ricardo Alvarez	ST	Argentina, Uruguay	1	0	0	0	NULL	NULL	NULL
12	Michel Dubois	LW, RW	France	0	0	0	0	NULL	NULL	NULL

```
s.scout_id,
s.scout_name,
GROUP_CONCAT(DISTINCT pos.position_name ORDER BY pos.position_name SEPARATOR ',
    ') AS expertise_positions,
GROUP_CONCAT(DISTINCT c.country_name ORDER BY c.country_name SEPARATOR ', ') AS
    expertise_countries,
COUNT(DISTINCT sr.report_id) AS total_reports,
SUM(CASE WHEN sr.recommendation = 'Sign' THEN 1 ELSE 0 END) AS
    sign_recommendations,
SUM(CASE WHEN sr.was_signed = 1 THEN 1 ELSE 0 END) AS players_signed,
SUM(CASE WHEN sr.signing_successful = 1 THEN 1 ELSE 0 END) AS successful_signings,
ROUND(
  (SUM(CASE WHEN sr.was_signed = 1 THEN 1 ELSE 0 END) * 100.0) /
  NULLIF(SUM(CASE WHEN sr.recommendation = 'Sign' THEN 1 ELSE 0 END), 0),
  2
) AS signing_conversion_rate_pct,
ROUND(
  (SUM(CASE WHEN sr.signing_successful = 1 THEN 1 ELSE 0 END) * 100.0) /
  NULLIF(SUM(CASE WHEN sr.was_signed = 1 THEN 1 ELSE 0 END), 0),
    2
  ) AS success_rate_pct,
  ROUND(AVG(CASE WHEN sr.was_signed = 1 THEN perf.average_rating
    END), 2) AS avg_rating_of_signed_players
FROM Scout s
LEFT JOIN Scout_Expertise se ON s.scout_id = se.scout_id
LEFT JOIN Position pos ON se.position_id = pos.position_id
LEFT JOIN Country c ON se.country_id = c.country_id
LEFT JOIN Scout_Report sr ON s.scout_id = sr.scout_id
LEFT JOIN Performance perf ON sr.player_id = perf.player_id AND
    perf.season = '2023-2024'
GROUP BY s.scout_id, s.scout_name
ORDER BY success_rate_pct DESC, signing_conversion_rate_pct DESC;
```

5. Global Marketability and Fanbase Reach Analysis

- **Description:** List players with social media followings above 500K AND performance ratings above 7.0, showing their follower growth rate over the past year, whether they've represented their national team in major tournaments (World Cup, continental championships), whether they play in a league different from their home country, and their home country's , grouped by age and position.

player_id	player_name	age	primary_position	current_club	nationality	club_country	plays_abroad	avg_performance_rating	total_followers	avg_growth_rate_pct	national_team_appearances	tournaments	major_tournament_participant	market_value_millions_eur
4	Jean Dubois	26	CAM	Paris Saint-Germain	France	France	No	8.10	9400000	16.55	1	World Cup	Yes	65.00
11	Bruno Fernandes	25	CAM	Real Madrid	Portugal	Spain	Yes	8.60	6600000	22.70	1	Continental Championship	Yes	95.00
3	Diego Martinez	21	LW	Liverpool	Argentina	England	Yes	7.90	5790000	29.00	1	Continental Championship	Yes	45.00
7	Rafael Costa	27	RW	Porto	Portugal	Portugal	No	8.50	4000000	18.40	1	Continental Championship	Yes	75.00
1	Marcus Silva	23	ST	Flamengo	Brazil	Brazil	No	7.87	3810000	13.90	1	Continental Championship	Yes	32.00
2	Luca Vandenberg	24	CM	Club Brugge	Belgium	Belgium	No	7.90	960000	8.70	1	Continental Championship	Yes	25.00
6	Takeshi Yamamoto	25	DM	Yokohama F. Marinos	Japan	Japan	No	7.70	900000	11.80	1	Continental Championship	Yes	22.00
8	Ahmed El-Sayed	22	RB	Manchester United	England	England	No	7.70	620000	9.40	1	World Cup	Yes	35.00

```
SELECT
  p.player_id,
  CONCAT(p.first_name, ' ', p.last_name) AS player_name,
  TIMESTAMPDIFF(YEAR, p.date_of_birth, CURDATE()) AS age,
  pos.position_name AS primary_position,
  cl.club_name AS current_club,
  home_country.country_name AS nationality,
  club_country.country_name AS club_country,
  CASE
    WHEN home_country.country_id != club_country.country_id THEN 'Yes'
    ELSE 'No'
  END AS plays_abroad,
  ROUND(AVG(perf.average_rating), 2) AS avg_performance_rating,
  SUM(sm.follower_count) AS total_followers,
  ROUND(AVG(sm.growth_rate), 2) AS avg_growth_rate_pct,
  COUNT(DISTINCT nta.appearance_id) AS national_team_appearances,
  GROUP_CONCAT(DISTINCT nta.tournament_type ORDER BY nta.tournament_year DESC
    SEPARATOR ', ') AS tournaments,
  MAX(CASE WHEN nta.tournament_type IN ('World Cup', 'Continental Championship') THEN 'Yes' ELSE
    'No' END) AS major_tournament_participant,
  ROUND(mv.valuation / 1000000, 2) AS market_value_millions_eur
```

```
SELECT
  p.player_id,
  CONCAT(p.first_name, ' ', p.last_name) AS player_name,
  TIMESTAMPDIFF(YEAR, p.date_of_birth, CURDATE()) AS age,
  pos.position_name AS primary_position,
  cl.club_name AS current_club,
  home_country.country_name AS nationality,
  club_country.country_name AS club_country,
  CASE
    WHEN home_country.country_id != club_country.country_id
THEN 'Yes'
    ELSE 'No'
  END AS plays_abroad,
  ROUND(AVG(perf.average_rating), 2) AS
avg_performance_rating,
  SUM(sm.follower_count) AS total_followers,
  ROUND(AVG(sm.growth_rate), 2) AS avg_growth_rate_pct,
  COUNT(DISTINCT nta.appearance_id) AS
national_team_appearances,
  GROUP_CONCAT(DISTINCT nta.tournament_type ORDER BY
nta.tournament_year DESC SEPARATOR ', ') AS tournaments,
  MAX(CASE WHEN nta.tournament_type IN ('World Cup',
'Continental Championship') THEN 'Yes' ELSE 'No' END) AS
major_tournament_participant,
  ROUND(mv.valuation / 1000000, 2) AS market_value_millions_eur
```

6. Agent Portfolio Performance and Negotiation Patterns

- **Description:** For each agent representing 5 or more players in the database, calculate the average transfer fee their clients received compared to similar players (same position, age range, and performance rating) represented by other agents, and identify agents whose clients have the highest percentage of successful transfers (increase in player value after transfer).

agent_id	agent_name	total_clients	total_transfers	avg_fee_millions_eur	market_avg_millions_eur	premium_over_market_pct	avg_client_rating	successful_value_increases	success_rate_pct
11	César Boaventura	2	2	25.25	20.02	26.12	8.36	0	0.00
10	Ulises Forte	4	1	15.00	20.02	-25.07	8.03	2	200.00
12	Rodger Linse	2	2	3.86	20.02	-80.72	7.90	0	0.00

```
WITH AgentStats AS (
  SELECT
    a.agent_id,
    a.agent_name,
    COUNT(DISTINCT p.player_id) AS total_clients,
    COUNT(DISTINCT t.transfer_id) AS total_transfers,
    ROUND(AVG(t.transfer_fee), 2) AS avg_transfer_fee,
    ROUND(AVG(perf.average_rating), 2) AS avg_client_rating,
    SUM(CASE
      WHEN mv_after.valuation > mv_before.valuation THEN 1
      ELSE 0
    END) AS successful_value_increases
  FROM Agent a
  JOIN Player p ON a.agent_id = p.agent_id
  LEFT JOIN Transfer t ON p.player_id = t.player_id AND t.transfer_type = 'Permanent'
  LEFT JOIN Performance perf ON p.player_id = perf.player_id AND perf.season = '2023-
    2024'
  LEFT JOIN Market_Value mv_before ON p.player_id = mv_before.player_id
    AND mv_before.recorded_at < COALESCE(t.transfer_date, '2024-01-01')
  LEFT JOIN Market_Value mv_after ON p.player_id = mv_after.player_id
    AND mv_after.recorded_at > COALESCE(t.transfer_date, '2024-01-01')
  GROUP BY a.agent_id, a.agent_name
  HAVING COUNT(DISTINCT p.player_id) >= 2
),
MarketComparison AS (
  SELECT
    ROUND(AVG(t.transfer_fee), 2) AS market_avg_transfer_fee
  FROM Transfer t
  WHERE t.transfer_type = 'Permanent'
   AND t.transfer_fee IS NOT NULL
)
SELECT
  ast.agent_id,
  ast.agent_name,
  ast.total_clients,
  ast.total_transfers,
  ROUND(ast.avg_transfer_fee / 1000000, 2) AS
avg_fee_millions_eur,
  ROUND(mc.market_avg_transfer_fee / 1000000, 2) AS
market_avg_millions_eur,
  ROUND(((ast.avg_transfer_fee - mc.market_avg_transfer_fee) /
mc.market_avg_transfer_fee) * 100, 2) AS
premium_over_market_pct,
  ast.avg_client_rating,
  ast.successful_value_increases,
  ROUND((ast.successful_value_increases * 100.0) /
NULLIF(ast.total_transfers, 0), 2) AS success_rate_pct
FROM AgentStats ast
CROSS JOIN MarketComparison mc
ORDER BY premium_over_market_pct DESC;
```

7. Contract Expiration Opportunity Window

- **Description:** Show players with contracts expiring within 12 months who have performance ratings above 7.5, grouped by position and current market value.

player_id	player_name	age	primary_position	position_category	current_club	league_name	nationality	contract_end_date	days_until_expiry	months_until_expiry	avg_performance_rating	current_value_millions_eur	estimated_reduced_fee_millions	agent_name	agent_contact
7	Rafael Costa	27	RW	Attacker	Porto	Primeira Liga	Portugal	2026-06-30	211	7.0	8.50	75.00	37.50	César Boaventura	cboaventura@prosoccer.pt
5	Nikola Petrovic	22	CB	Defender	Dinamo Zagreb	Croatian First League	Croatia	2026-06-30	211	7.0	8.00	28.00	14.00	Federico Pastorello	fpastorello@wp8.com
12	André Oliveira	22	LB	Defender	Ajax Amsterdam	Eredivisie	Portugal	2026-06-30	211	7.0	7.80	12.00	6.00	César Boaventura	cboaventura@prosoccer.pt
8	Ahmed El-Sayed	22	RB	Defender	Manchester United	Premier League	England	2026-06-30	211	7.0	7.70	35.00	17.50	Jonathan Barnett	jbarnett@stellar-group.com

```
SELECT
  p.player_id,
  CONCAT(p.first_name, ' ', p.last_name) AS player_name,
  TIMESTAMPDIFF(YEAR, p.date_of_birth, CURDATE()) AS age,
  pos.position_name AS primary_position,
  pos.position_category,
  cl.club_name AS current_club,
  l.league_name,
  c.country_name AS nationality,
  t.contract_end_date,
  DATEDIFF(t.contract_end_date, CURDATE()) AS days_until_expiry,
  ROUND(DATEDIFF(t.contract_end_date, CURDATE()) / 30, 1) AS months_until_expiry,
  ROUND(AVG(perf.average_rating), 2) AS avg_performance_rating,
  ROUND(mv.valuation / 1000000, 2) AS current_value_millions_eur,
  ROUND(mv.valuation * 0.5 / 1000000, 2) AS estimated_reduced_fee_millions,
  a.agent_name,
  a.email AS agent_contact
FROM Player p
JOIN Position pos ON p.primary_position_id = pos.position_id
JOIN Club cl ON p.current_club_id = cl.club_id
JOIN Country c ON p.nationality_id = c.country_id
JOIN Transfer t ON p.player_id = t.player_id
AND t.transfer_id = (SELECT MAX(transfer_id) FROM Transfer WHERE player_id =
    p.player_id)
JOIN Performance perf ON p.player_id = perf.player_id AND perf.season = '2023-2024'
LEFT JOIN Market_Value mv ON p.player_id = mv.player_id
  AND mv.recorded_at = (SELECT MAX(recorded_at) FROM Market_Value WHERE
player_id = p.player_id)
LEFT JOIN Agent a ON p.agent_id = a.agent_id
WHERE t.contract_end_date <= DATE_ADD(CURDATE(), INTERVAL 12 MONTH)
 AND t.contract_end_date >= CURDATE()
GROUP BY p.player_id, p.first_name, p.last_name, p.date_of_birth,
pos.position_name,
    pos.position_category, cl.club_name, l.league_name, c.country_name,
    t.contract_end_date, mv.valuation, a.agent_name, a.email
HAVING avg_performance_rating > 7.5
ORDER BY months_until_expiry ASC, avg_performance_rating DESC;
```

8. Scout Coverage Gaps

- **Description:** Identify countries and leagues that have players meeting performance thresholds (rating >7.0, age <28) but have been scouted fewer than 3 times in the past year.

country_name	league_name	ranking_tier	quality_players_count	avg_rating	avg_age	scout_reports_last_year	unique_scouts_assigned	last_scouted_date	days_since_last_scout	priority_status	players_per_scout_report	top_players
England	Premier League	1	2	7.97	21.3	0	0	NULL	NULL	CRITICAL - No Scouting Activity	NULL	Diego Martinez, Ahme
Spain	La Liga	1	1	8.60	25.0	0	0	NULL	NULL	CRITICAL - No Scouting Activity	NULL	Bruno Fernandes
Portugal	Primeira Liga	2	1	8.50	27.0	0	0	NULL	NULL	CRITICAL - No Scouting Activity	NULL	Rafael Costa
France	Ligue 1	1	1	8.30	26.0	0	0	NULL	NULL	CRITICAL - No Scouting Activity	NULL	Jean Dubois
Germany	Bundesliga	1	1	8.20	26.0	0	0	NULL	NULL	CRITICAL - No Scouting Activity	NULL	Lars Hoffman
Brazil	Brasileirão	2	1	8.00	23.0	0	0	NULL	NULL	CRITICAL - No Scouting Activity	NULL	Marcus Silva
Croatia	Croatian First League	3	1	8.00	22.0	0	0	NULL	NULL	CRITICAL - No Scouting Activity	NULL	Nikola Petrovic
Belgium	Belgian Pro League	2	1	7.90	24.0	0	0	NULL	NULL	CRITICAL - No Scouting Activity	NULL	Luca Vandenberg
Mexico	Liga MX	2	1	7.90	24.0	0	0	NULL	NULL	CRITICAL - No Scouting Activity	NULL	Mateo Gonzalez
Japan	J1 League	2	1	7.80	25.0	0	0	NULL	NULL	CRITICAL - No Scouting Activity	NULL	Takeshi Yamamoto
Netherlands	Eredivisie	2	1	7.80	22.0	0	0	NULL	NULL	CRITICAL - No Scouting Activity	NULL	André Oliveira

```
SELECT
  c.country_name,
  l.league_name,
  l.ranking_tier,
  COUNT(DISTINCT p.player_id) AS quality_players_count,
  ROUND(AVG(perf.average_rating), 2) AS avg_rating,
  ROUND(AVG(TIMESTAMPDIFF(YEAR, p.date_of_birth, CURDATE())), 1) AS
    avg_age,
  COUNT(DISTINCT sr.report_id) AS scout_reports_last_year,
  COUNT(DISTINCT sr.scout_id) AS unique_scouts_assigned,
  MAX(sr.report_date) AS last_scouted_date,
  DATEDIFF(CURDATE(), MAX(sr.report_date)) AS days_since_last_scout,
  CASE
    WHEN COUNT(DISTINCT sr.report_id) = 0 THEN 'CRITICAL - No Scouting
    Activity'
    WHEN COUNT(DISTINCT sr.report_id) = 1 THEN 'HIGH PRIORITY - Minimal
    Coverage'
    WHEN COUNT(DISTINCT sr.report_id) = 2 THEN 'MEDIUM PRIORITY - Limited
    Coverage'
    ELSE 'Adequate Coverage'
  END AS priority_status,
ROUND(COUNT(DISTINCT p.player_id) / NULLIF(COUNT(DISTINCT
sr.report_id), 0), 2) AS players_per_scout_report,
  GROUP_CONCAT(DISTINCT CONCAT(p.first_name, ' ', p.last_name)
ORDER BY perf.average_rating DESC SEPARATOR ', ') AS top_players
FROM Country c
JOIN League l ON c.country_id = l.country_id
JOIN Club cl ON l.league_id = cl.league_id
JOIN Player p ON cl.club_id = p.current_club_id
JOIN Performance perf ON p.player_id = perf.player_id
LEFT JOIN Scout_Report sr ON p.player_id = sr.player_id
  AND sr.report_date >= DATE_SUB(CURDATE(), INTERVAL 1 YEAR)
WHERE perf.average_rating > 7.0
 AND TIMESTAMPDIFF(YEAR, p.date_of_birth, CURDATE()) < 28
 AND perf.season = '2023-2024'
GROUP BY c.country_name, l.league_name, l.ranking_tier
HAVING quality_players_count >= 1
 AND scout_reports_last_year < 3
ORDER BY
  scout_reports_last_year ASC,
  quality_players_count DESC,
  avg_rating DESC;
```

Challenges

- We had issues forward engineering.
- Using Class table(Supertype/Subtype) approach made ERD large complex.
- Handling errors while creating large tables with many constraints.
- Writing long SQL queries that required multiple joins

Lessons

- Always try and manage scope.
- Planning the structure early prevents confusion later.
- Clean and consistent data is more important than large data.
- Complex queries become easy once relationships are clear.

Any Questions?

Thank You!